FLAGSHIP 03 · SMART ERGONOMIC SENSING SERIES

LC Gloves-01

Wrist Repetitive Work Assessment Glove · Edge-AI Biometric Diagnostic Glov

Edge-AI Biometric Glove. GBZ 336-2025 Carpal Tunnel Risk Diagnostics.

Designed for manufacturing jobs with repetitive forceful wrist movements, the LC Gloves-01 strictly complies with GBZ 336-2025 Diagnostic Criteria for Occupational Carpal Tunnel Syndrome. Equipped with dual-posture IMU sensors (dorsal hand + wrist cuff) and onboard edge-AI computing, it tracks wrist angles, valid work motions, operation frequency, duration, and load intensity in real time. Supports wireless data upload and tablet analytics for ergonomic risk quantification and workstation optimization.

01 Dual-Point Attitude Sensing

Differential sensing architecture cancels whole-arm movement artifacts, isolating pure wrist flexion, extension, and deviation.

02 Onboard Edge-AI Processing

Glove micro-controller calculates angles, counts valid cycles, and compiles metrics locally with zero latency.

03 Wireless Tablet Telemetry

Real-time Bluetooth/Wi-Fi data stream to companion PC/tablet suite generates exportable diagnostic risk reports.

04 Industrial-Grade Anti-Interference

Shielded against electromagnetic noise and machine vibration in automotive and electronics plants.

LC Gloves-01 · OFFICIAL HARDWARE VIEW



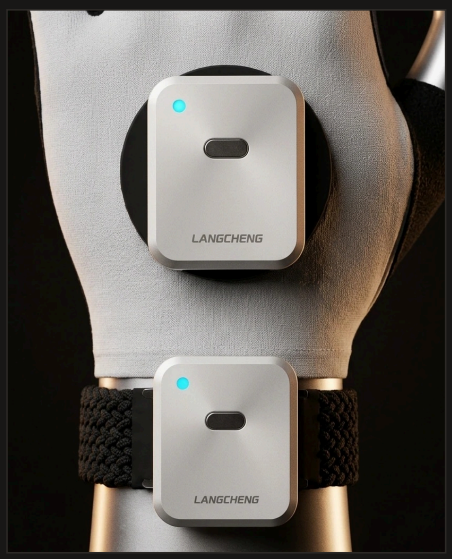
KEY METRIC 01
<= ±0.1°
Static IMU Measurement Precision

KEY METRIC 02
±180°
Full Multi-Axis Range of Motion

KEY METRIC 03
200g
Featherlight Ergonomic Form Factor

KEY METRIC 04
24/min
Automated Valid Motion Frequency Tracking

MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



DUAL DIFFERENTIAL IMU · · ±0.1° PRECISION

Dual-Posture Differential IMU Sensor Pods

By pairing a dorsal metacarpal IMU with a wrist-cuff reference IMU, the onboard edge-AI processor rejects gross arm motion and isolates true wrist flexion/extension and radial/ulnar deviation past the 15° recognition threshold.



TECHNICAL GREY ANTI-SLIP TEXTILE

High-Stretch Technical Knit & Half-Finger Cutouts

200g breathable technical grey textile keeps IMU sensors locked without skin migration while leaving fingertips free for micro-fasteners and soldering.



GBZ 336-2025 COMPLIANT ANALYTICS

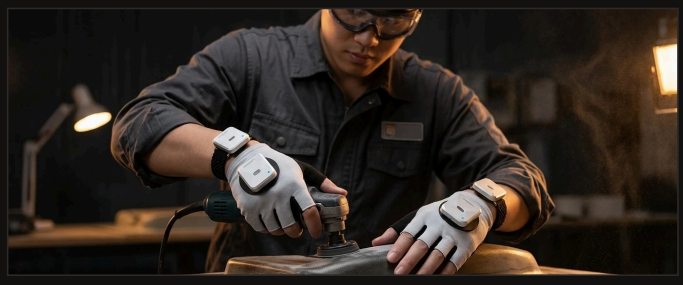
GBZ 336-2025 Diagnostic Waveform Analytics Suite

Streams real-time wrist kinematics to an industrial tablet PC to log daily working hours, valid motion counts, and automated occupational carpal tunnel syndrome risk reports.

ANATOMICAL & MECHANICAL SUBSYSTEMS (5 NODES)

1. **Half-Finger Tactile Cutouts [Distal Phalangeal Cutouts]**
   Exposed fingertips preserve fine touch for delicate assembly of miniature screws and electronics.
2. **Zero-Slack Dorsal Pod Mount [Primary Metacarpal IMU Node]**
   High-stretch elastic fabric locks the primary IMU sensor firmly over the metacarpals.
3. **Lightweight 200g Architecture [Breathable Technical Knit Chassis]**
   Minimizes muscle load during continuous 8-hour ergonomic audits.
4. **Adjustable Secondary Wristband [Radiocarpal Reference Strap]**
   Anchors the reference IMU sensor across varied wrist circumferences.
5. **Companion Diagnostic Software & Reference IMU [Wireless Edge-AI Telemetry Link]**
   Secondary wrist IMU pod streams real-time dual-channel waveforms for flexion/extension, radial/ulnar deviation, and GBZ 336-2025 risk grading.

FIELD DEPLOYMENT | 3C ELECTRONICS MANUFACTURING LINE | ERGONOMIC AUDIT STATION



Precision Electronics Assembly & Occupational Health Audits

Occupational disease prevention hospitals, third-party ergonomic testing agencies, and manufacturing EHS teams deploy the LC Gloves-01 directly on live production lines. As operators assemble precision components, the glove captures every flexion, extension, and ulnar deviation above the customizable 15° threshold - delivering legally defensible GBZ 336-2025 compliance reports.

GBZ 336-2025 Compliant	<= ±0.1° (±180° Range)	15° / 24 Motions min/min
Diagnostic Standard	Static Angle Accuracy	Standard Motion Threshold

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Gloves-01 (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Gloves-01
IDENTITY & COMPLIANCE	**Color**	Technical Light Grey with Matte White Sensor Pods & Black Trim
IDENTITY & COMPLIANCE	**Compliance**	GBZ 336-2025 Diagnostic Criteria for Occupational Carpal Tunnel Syndrome
IDENTITY & COMPLIANCE	**Applications**	Occupational disease prevention hospitals, third-party testing agencies, manufacturing workstation audits
PERFORMANCE & MECHANICS	**Recognition Threshold**	15° (customizable)
PERFORMANCE & MECHANICS	**Angle Range**	±180°
PERFORMANCE & MECHANICS	**Static Accuracy**	<= ±0.1°
PERFORMANCE & MECHANICS	**Frequency Counting**	Automatically counts valid motions (standard: 24 motions/min)
PERFORMANCE & MECHANICS	**Duration Tracking**	Daily working hour accumulation
PERFORMANCE & MECHANICS	**Measurement Dimensions**	Wrist flexion/extension, radial/ulnar deviation angles
DIMENSIONS & WARRANTY	**Sizes**	M, L, XL (customizable)
DIMENSIONS & WARRANTY	**Net Weight**	200g
DIMENSIONS & WARRANTY	**Package Dimensions**	23 x 17 x 8cm
DIMENSIONS & WARRANTY	**Optional Accessories**	Industrial Tablet PC (with pre-installed analytics suite) / Action Camera
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Gloves-01

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware